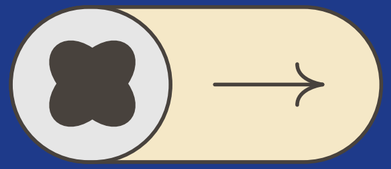
KAPLYSH OLHA &
SHOPSKA ANASTASIIA

XAI FOR CONTENT CREATORS



ONE MODEL DOESN'T FIT ALL CREATORS

PROBLEM

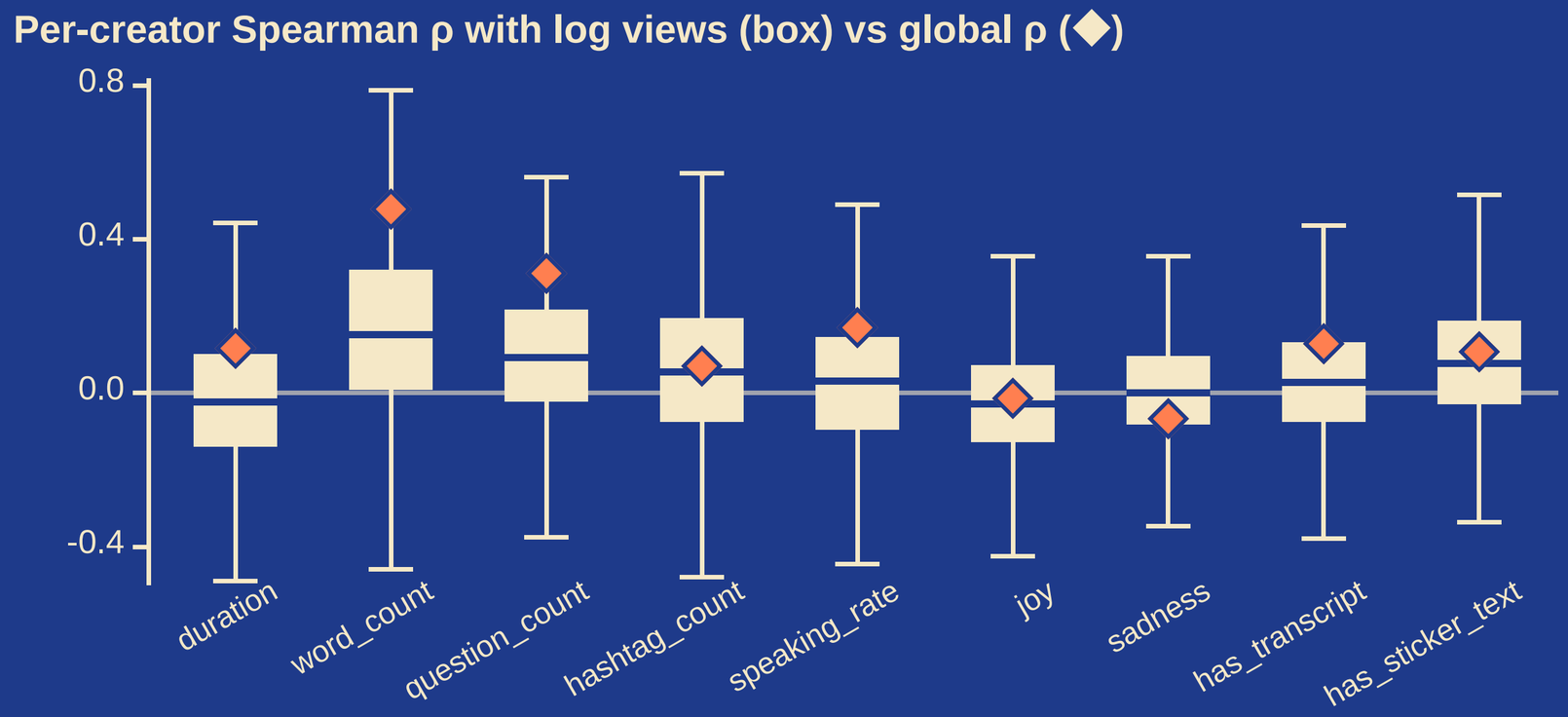
Existing tools like vidIQ, TubeBuddy and TikTok Creative Center give generic, unexplained advice.

HYPOTHESIS

Personalized explainable models beat generic ones.

EVIDENCE

Across features, 30–58% of creators show the opposite effect to the global trend.



DATASET

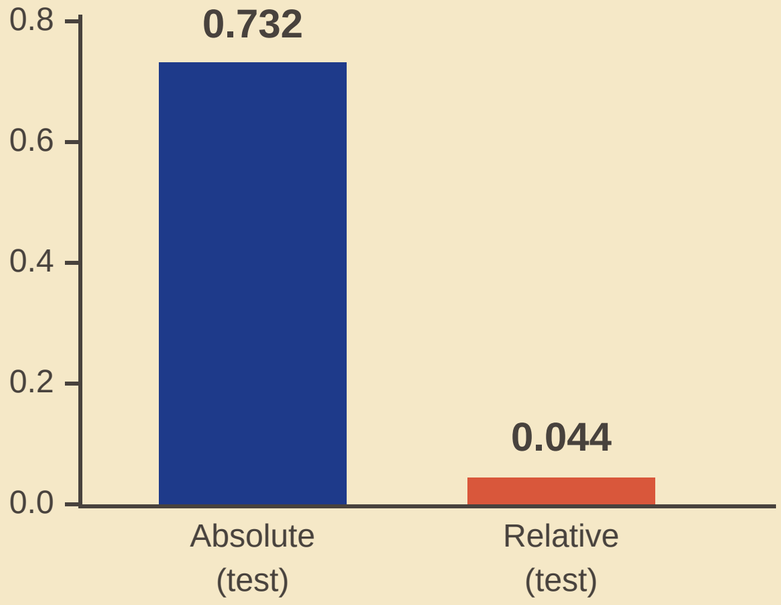
TikTok Video Engagement 200k (HuggingFace).
200,616 videos, 1,437 creators with ≥30 videos each.

TARGET

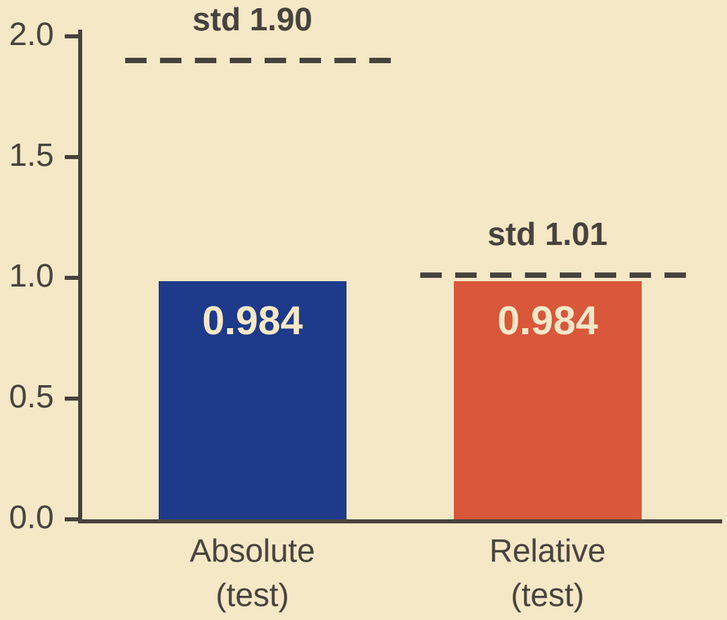
log(views) on day 14, relative to the creator's median.

Model "prediction = creator's train mean", test set

R^2: share of variance explained by creator



RMSE: typical prediction error



bars = RMSE of "creator only" model; dashed = std of target

WHY?

- **Why day 14:** 100% coverage, ~96% of final views already reached.
- **Why relative:** The creator explains 73% of absolute views and only 4% of relative ones, so the model learns from content.

PIPELINE

01 ✓
Raw data
3 HuggingFace tables

02 ✓
EDA
Exploring distributions, missing values and differences between creators to decide how to build the target and features

03 ✓
Feature engineering
106 features, leakage-safe

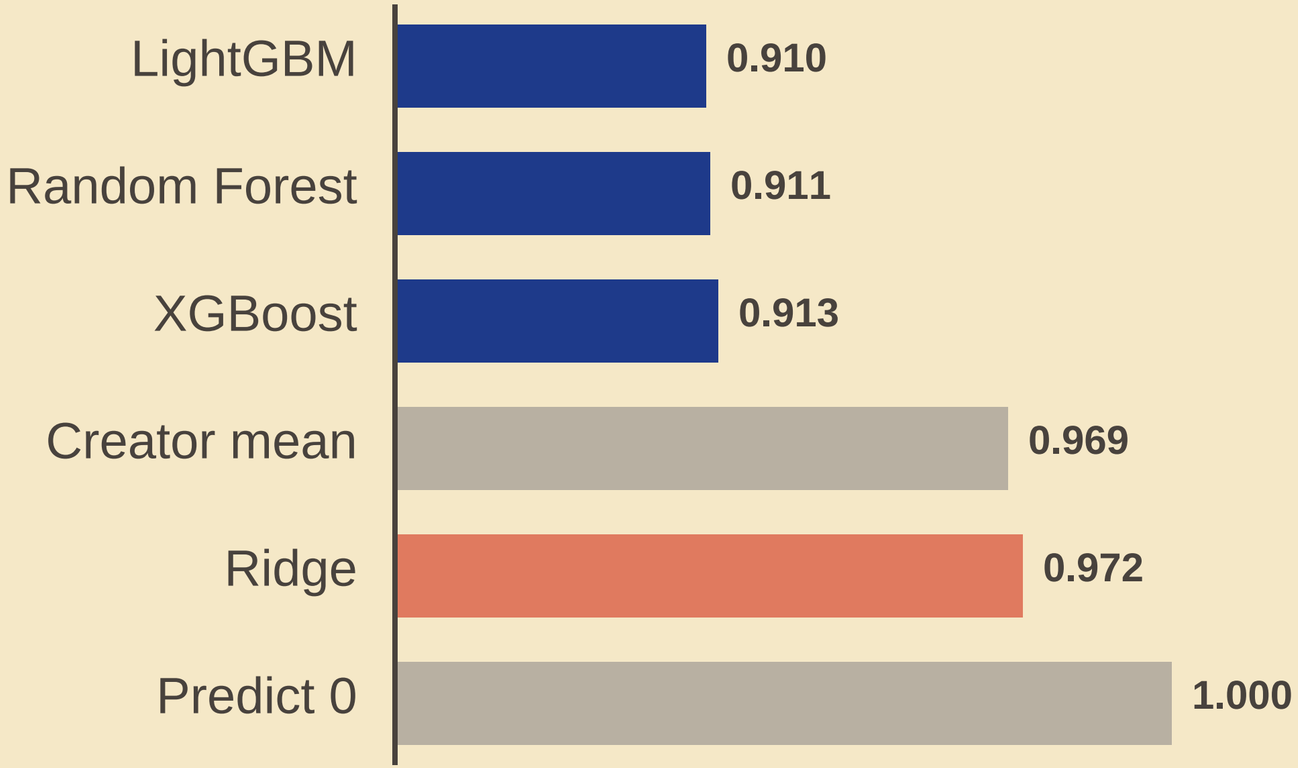
04 ✓
Global model
One model trained on all 1,437 creators: our baseline

05
Per-creator models
A separate model for each creator, trained on their own video history

06
SHAP + Recommendations
Explain what drives each prediction and turn it into concrete advice for the creator

✓ done faded = next steps

SETUP & RESULTS



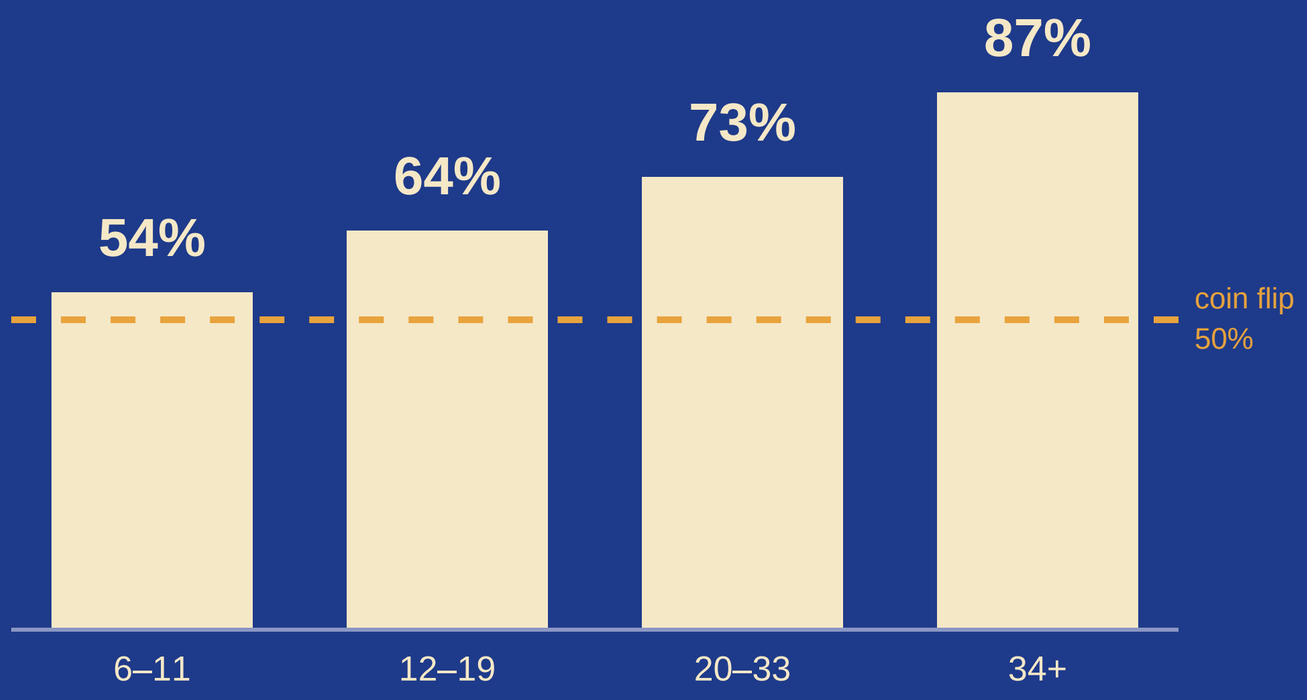
Test RMSE, lower is better (axis starts at 0.85)

- 4 models vs 2 trivial baselines
- Hyperparameters tuned with CV on train; test used once
- Metric: RMSE on relative target

- Tree models ≈ 0.91 RMSE: no clear winner.
- Ridge (0.972) loses even to creator mean (0.969), so non-linear models are needed.

WHO DOES THE MODEL HELP?

Share of creators where the model beats their own average

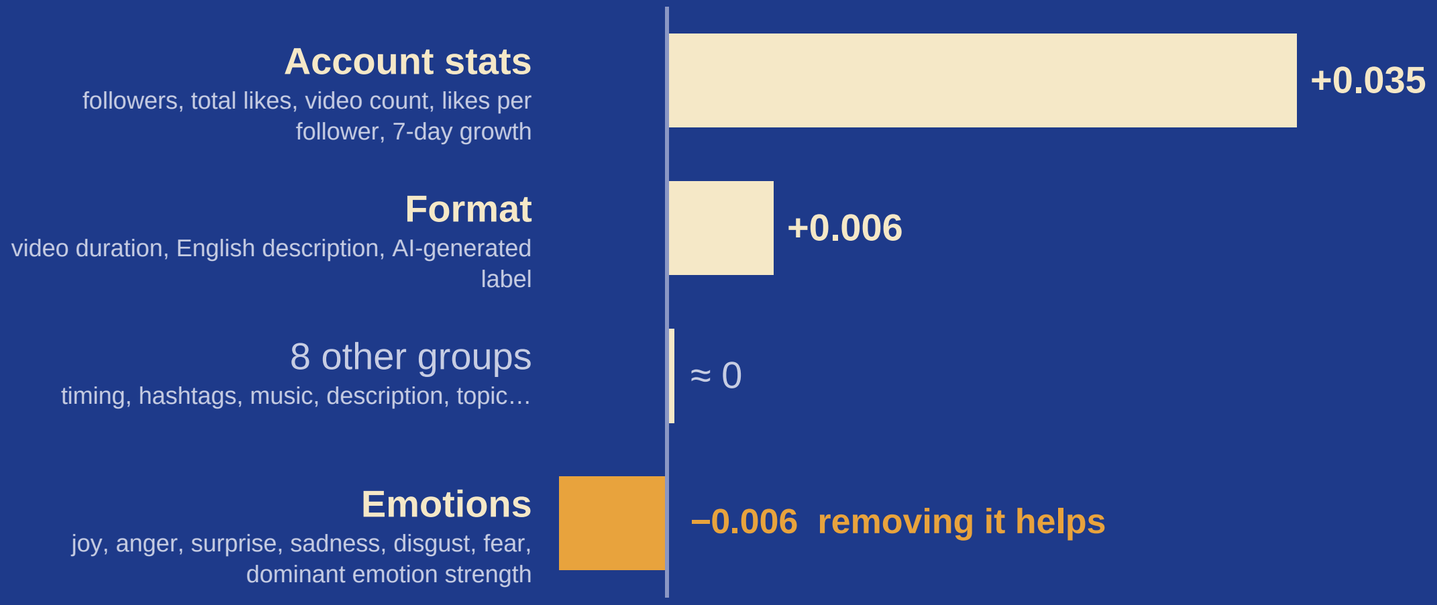


69% of creators: the model beats their own average
18%: it ranks their videos in reverse order

The global model works for creators with a long history, but for small creators it is close to a coin flip. One model doesn't fit everyone, which is why we move to personalization.

WHAT DRIVES VIEWS?

How much worse the model gets without each feature group (Δ CV RMSE)



Longer bar = more important group

- Account stats dominate: removing them hurts 6× more than any other group
- Format is the strongest group a creator controls
- Emotions only add noise: without them test RMSE drops 0.913 → 0.906

Main point: features the creator controls keep about a third of the model's gain over "same as usual".

THANK YOU

